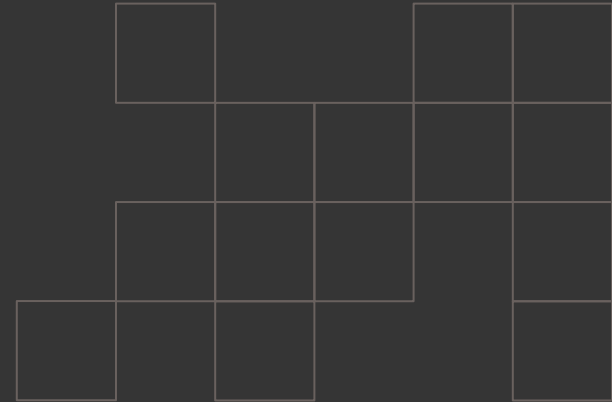


Haroon Amanullah
May 26, 2026

MMKF25 - Surface modelling, rendering and 3D

1. Non-Parametric Sparrow
2. Parametric Spaceship



Sparrow

Tools Used

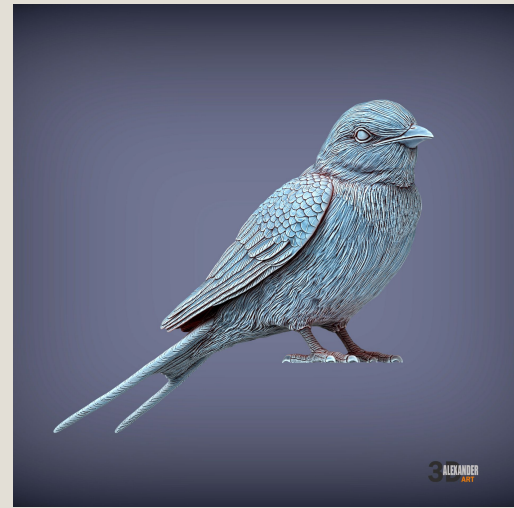
Blender 3D (Sculpting, Texturing, Rigging, Rendering)

Skills Learned

- Sculpting large and fine details
- Merging and blending objects into one surface
- UV unwrapping



Inspiration and Reference

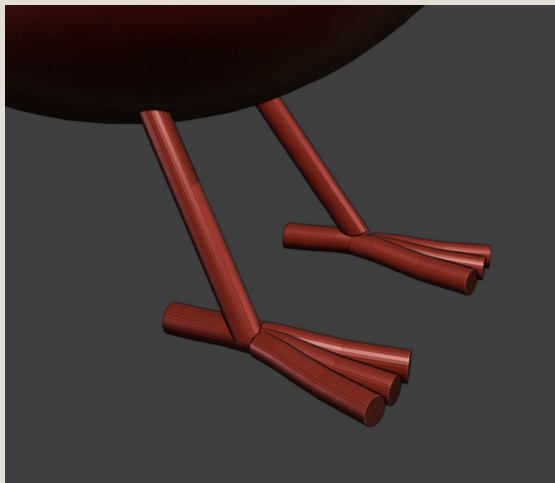


Sparrow — Process



Basic Sculpting

I started from a sphere and blocked out major shapes.



Arranging Primitives

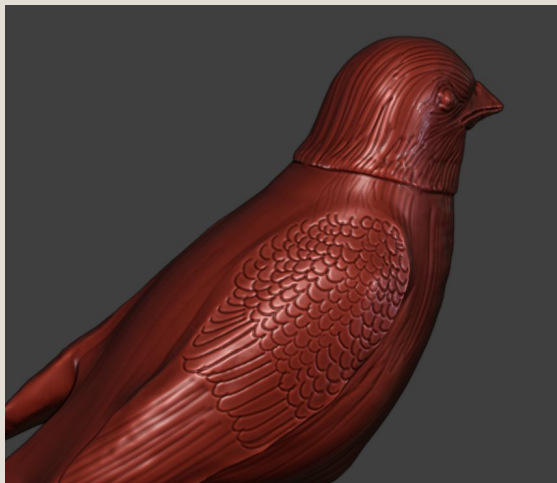
For more complex geometry like the feet, I blocked out shapes using cylinders.



Boolean Union + Sculpting

Then I used **Boolean Unions** to merge shapes and sculpt on top of it.

Sparrow — Process



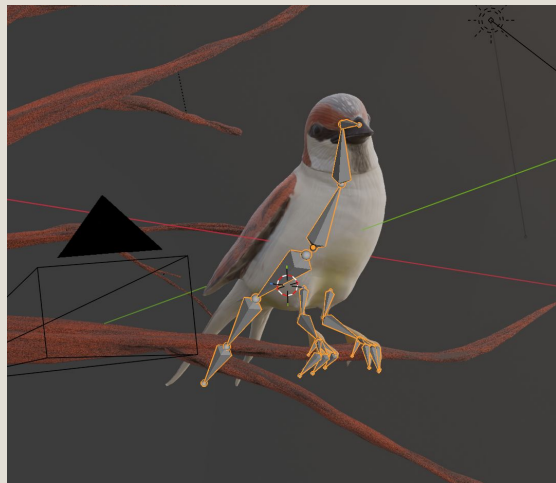
Sculpting Details

Remeshed at higher detail and **Hand-Sculpted** details.



UV Unwrapping + Texture Painting

Used default brushes to **Texture Paint** onto a 2k*2k image texture.



Rigging + Rendering

Matched an **Armature** to mesh, parented mesh with default weights, and posed it.

Spaceship

Tools Used

Solidworks - Surface Modeling

Blender - Rendering

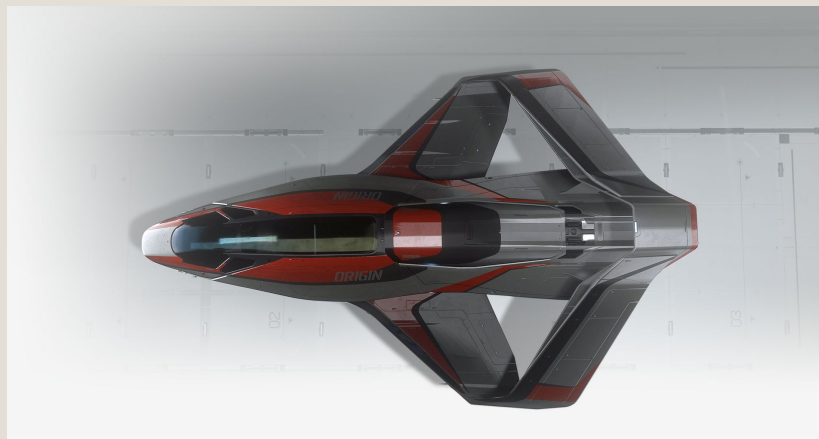
Skills Learned

- Modeling complex, flowing shapes using advanced surface modeling tools in Solidworks (Extrude, trim, loft, knit, revolve, offset)
- Applying multiple materials

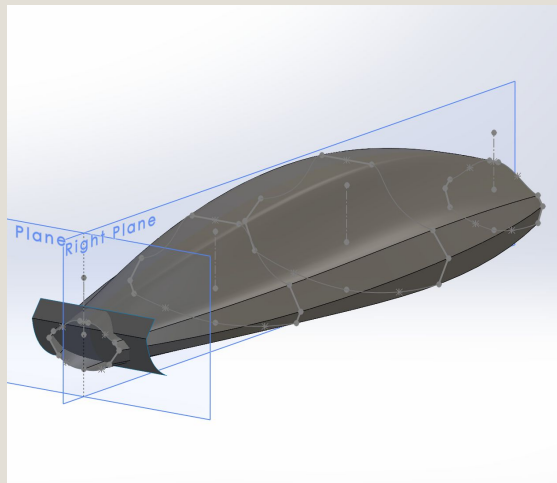


Inspiration and Reference

Origin 300i from *Star Citizen*
(video game)

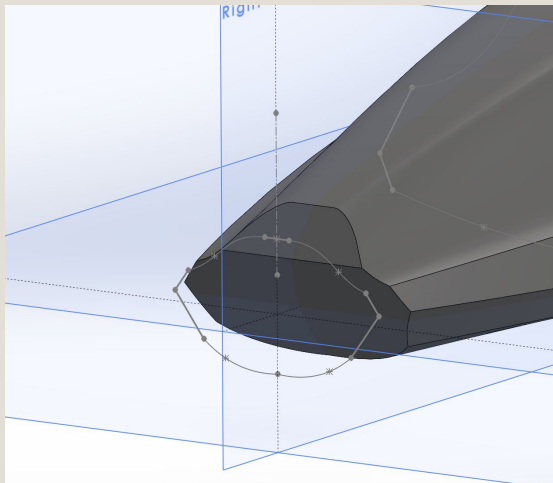


Spaceship — Process



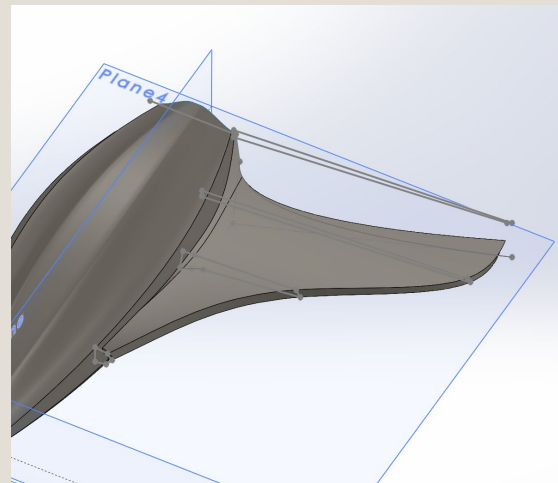
Lofting Curves

By **Lofting** 4 sketches I made the main body form.



Trimming and Knitting

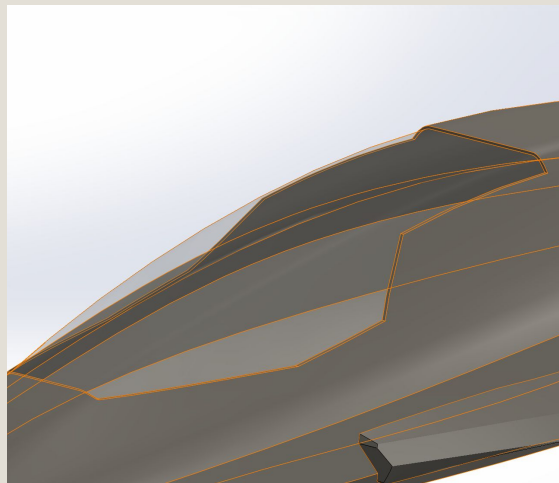
I used surface intersections from different angles to **Trim** and refine shapes before **Knitting** them together.



Creating and Merging Wing

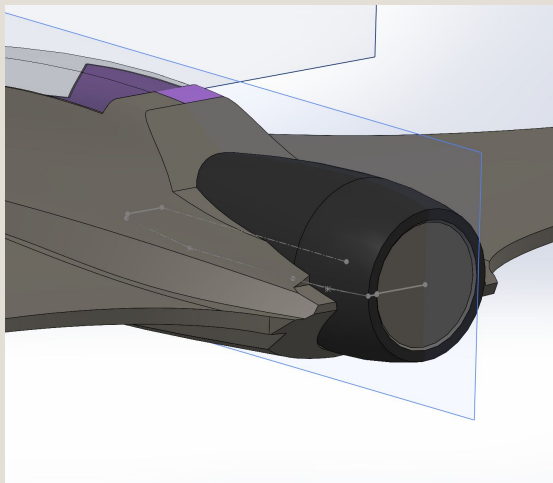
The wing was made similarly and then knit into 1 surface.

Spaceship — Process



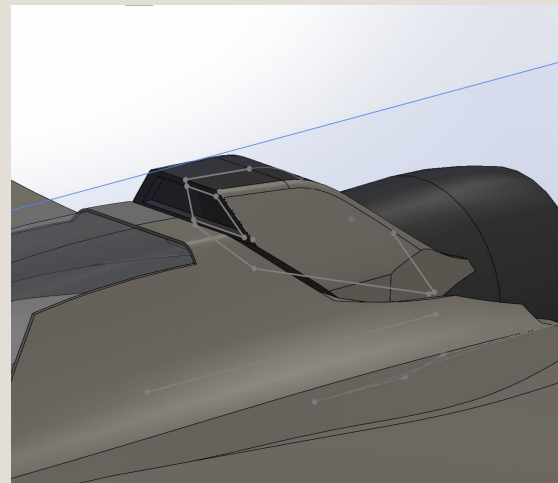
Insetting the Cockpit

I used the **Offset Surface** tool, and **Lofted** each edge together to knit into 1 surface.



Creating the Engine

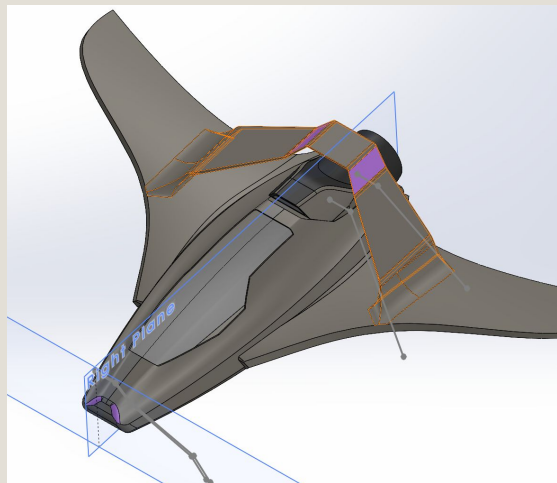
I used the **Revolved Surface** tool to create the engine from a profile sketch.



Sensor/Air Intake Module

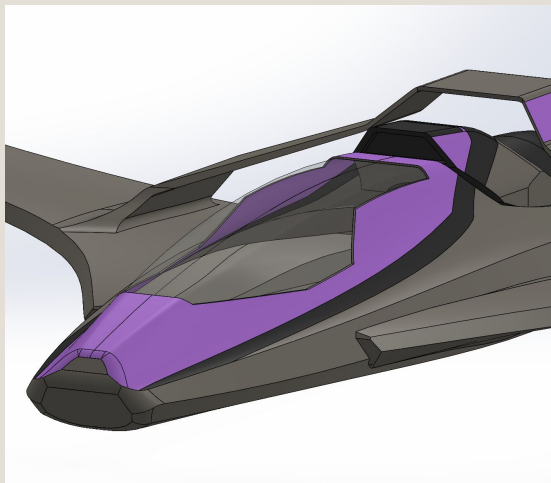
This challenging shape was made with 4 surfaces, 7 trims, an offset and fillet before being knit to the body.

Spaceship — Process



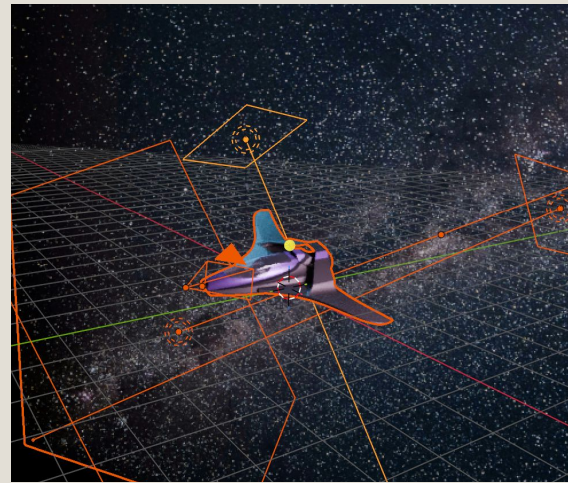
Upper Wing

I made this part using the intersection of 2 sketches and large **Fillets** at the bottom.



Creating a Paint Job

I used the **Split Line** feature to carve boundaries to assign various materials to.



Lighting and Rendering

Finally I exported as a **GLB** (includes material data) and rendered it in **Blender**.

Final Products

